EDA and MCMC Methods for Babies Dataset

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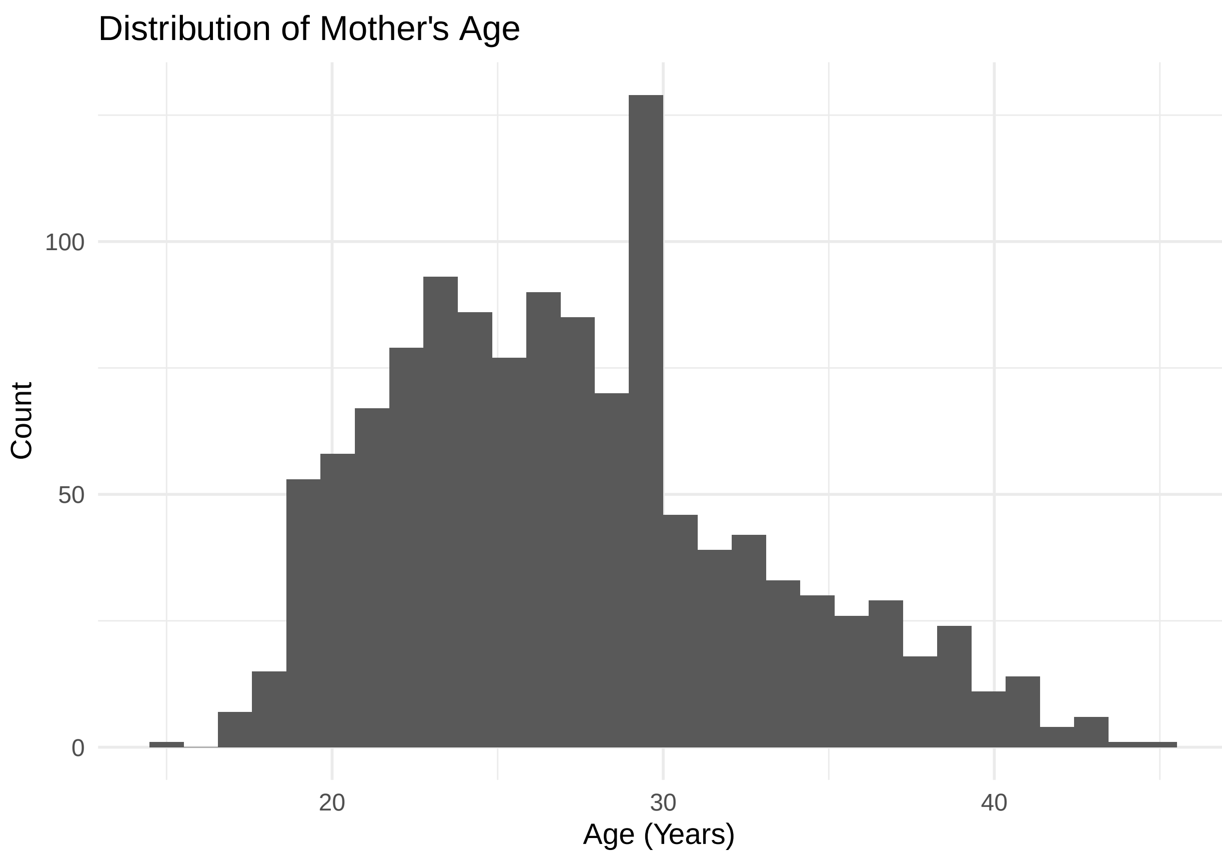
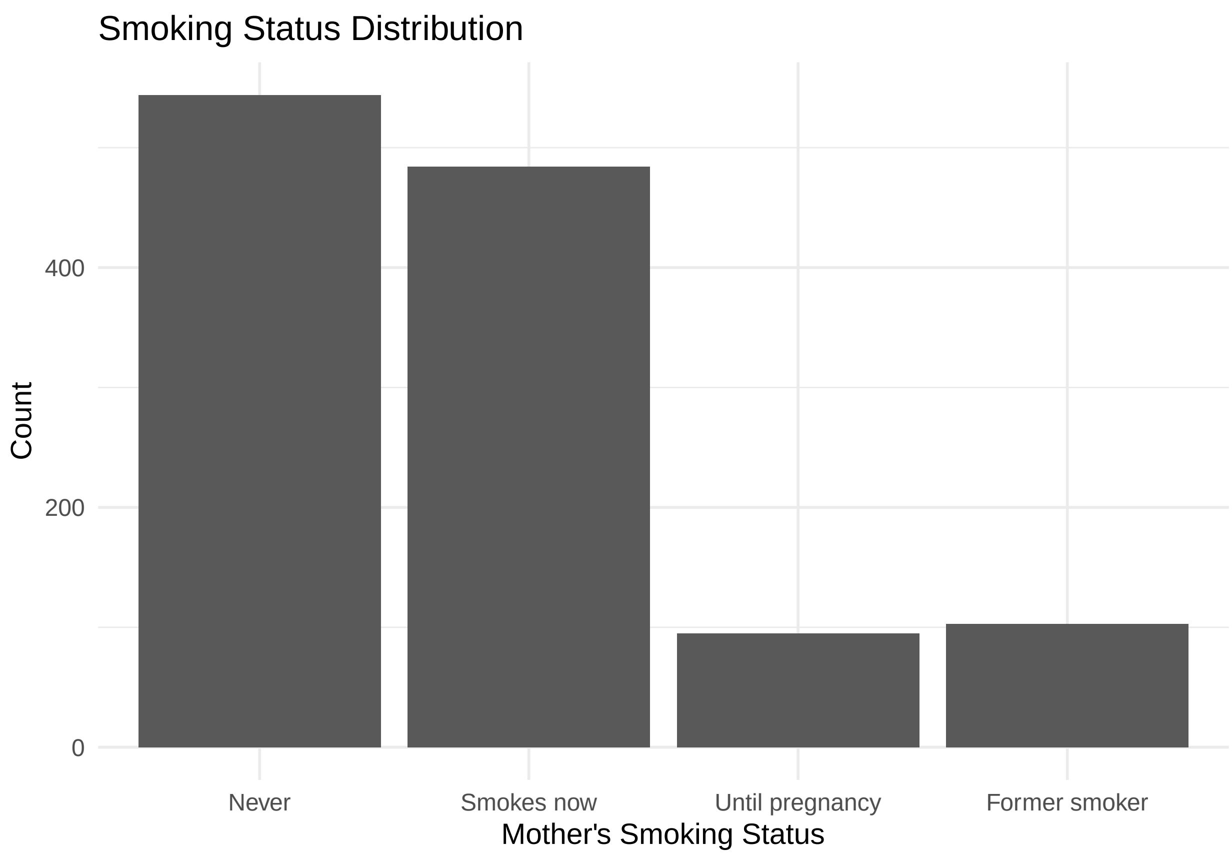
```r
library(UsingR)
library(Bolstad)
library(ggplot2)
library(reshape2)
library(ggridges)
```

EDA

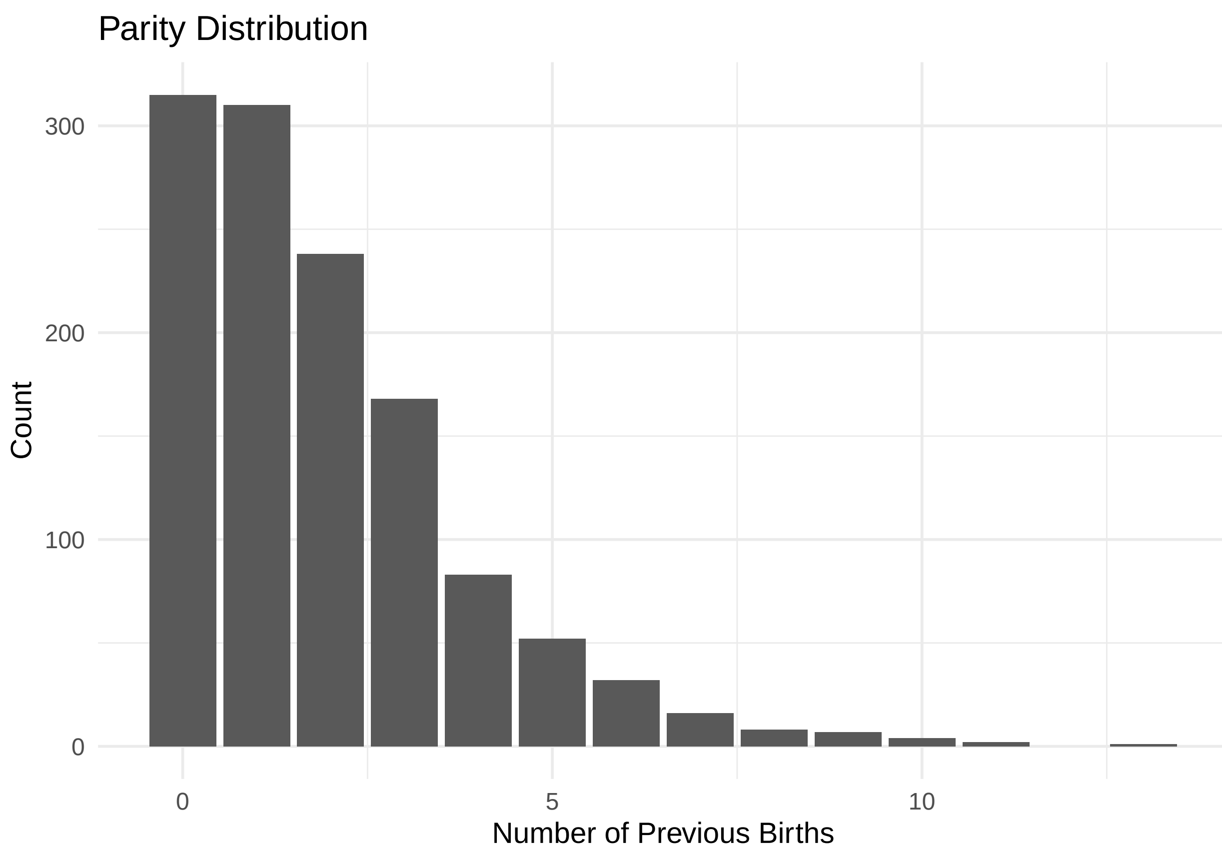
```r
babies_clean <- babies

#removing missing/incomplete data
babies_clean[babies_clean == 999] <- NA
babies_clean$age[babies_clean$age == 99] <- NA
babies_clean$wt1[babies_clean$wt1 == 999] <- NA
babies_clean$smoke[babies_clean$smoke == 9] <- NA
babies_clean$race[babies_clean$race == 99] <- NA
babies_clean$inc[babies_clean$inc == 99] <- NA
babies_clean$inc[babies_clean$inc == 98] <- NA
babies_clean$inc[babies_clean$sex == 9] <- NA
babies_clean$inc[babies_clean$ed == 9] <- NA
babies_clean$inc[babies_clean$ht == 99] <- NA

#age distribution
ggplot(babies_clean, aes(x = age)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Mother's Age",
       x = "Age (Years) ", y = "Count") +
  theme_minimal()
```



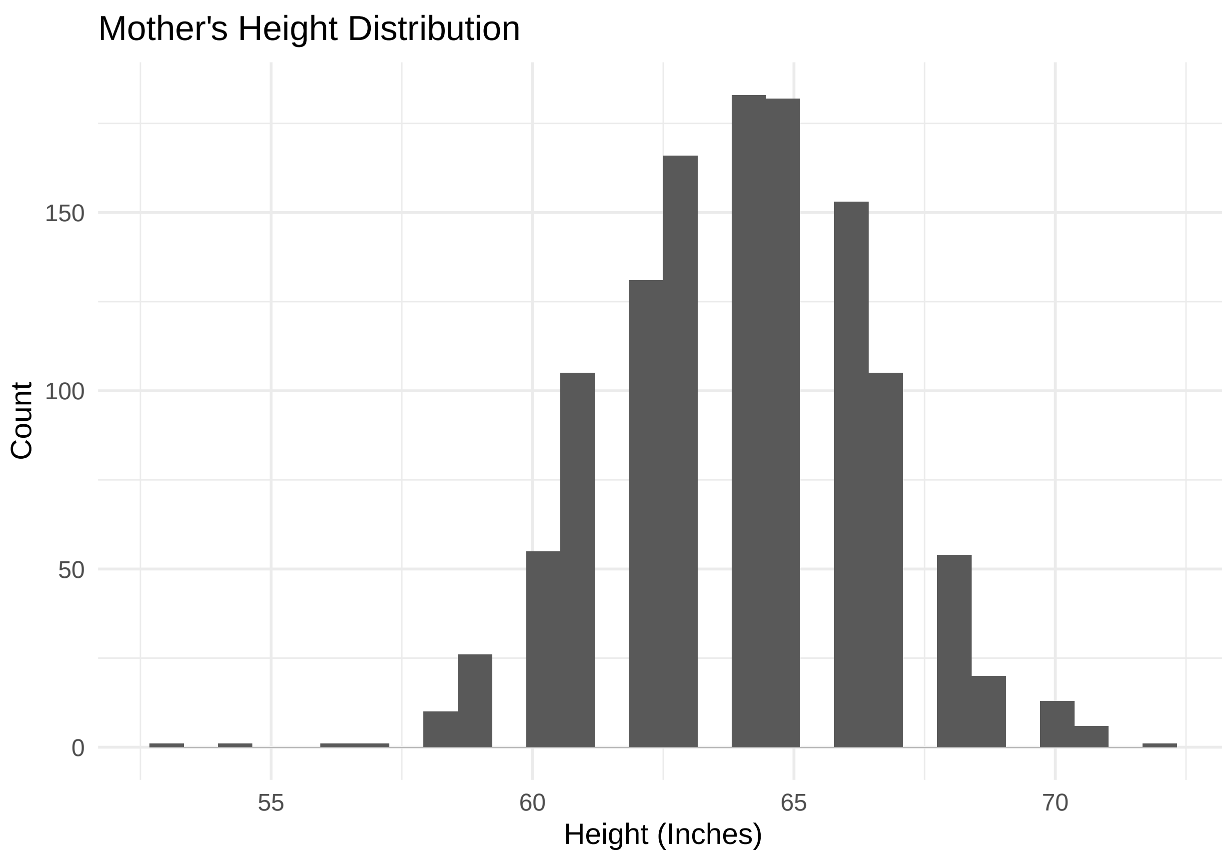

```r
#smoking status distributions
babies_clean$smoke_group <- with(babies_clean,
  ifelse(smoke == 0, "Never",
  ifelse(smoke == 1, "Smokes now",
  ifelse(smoke == 2, "Until pregnancy",
  ifelse(smoke == 3, "Former smoker", NA)))))

babies_clean$smoke_group <- factor(
  babies_clean$smoke_group,
  levels = c("Never", "Smokes now", "Until pregnancy", "Former smoker")
)

ggplot(subset(babies_clean, !is.na(smoke_group)), aes(x = smoke_group)) +
  geom_bar() +
  labs(
    title = "Smoking Status Distribution",
    x = "Mother's Smoking Status",
    y = "Count"
  ) +
  theme_minimal()
```



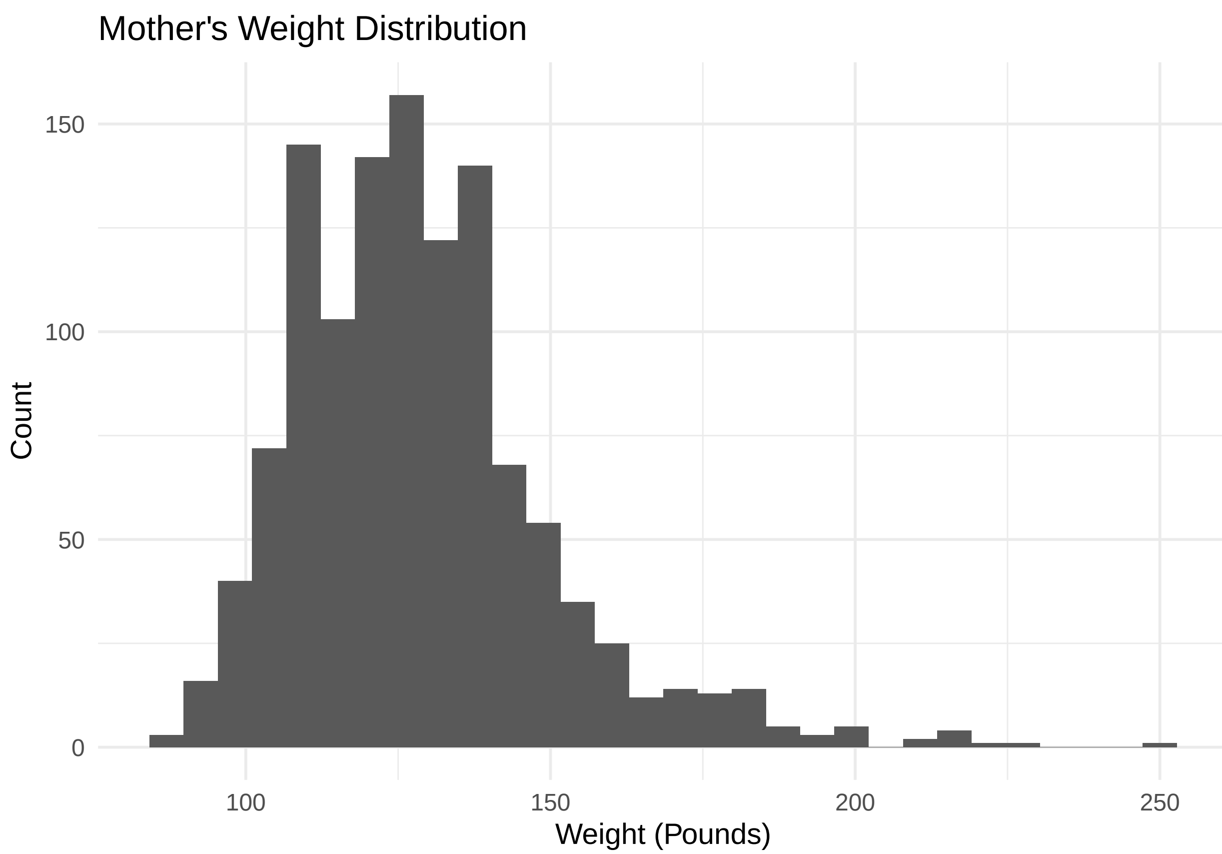

```
#parity distribution
ggplot(babies_clean, aes(x = parity)) +
  geom_bar() +
  labs(title = "Parity Distribution",
       x = "Number of Previous Births", y = "Count") +
  theme_minimal()
```



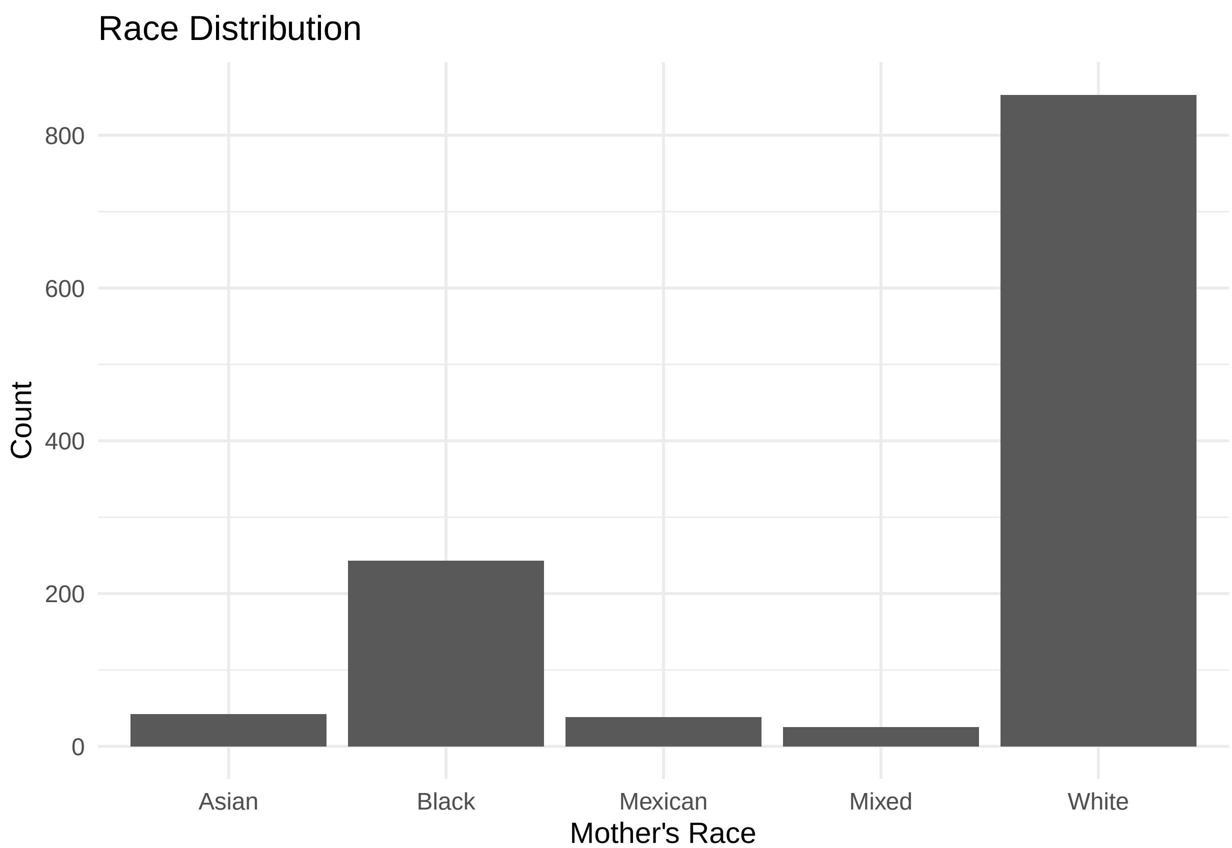

```r
#removing height outlier
babies_clean <- babies_clean[babies_clean$ht != 99, ]

#height distribution for moms
ggplot(babies_clean, aes(x = ht)) +
  geom_histogram(bins = 30) +
  labs(title = "Mother's Height Distribution",
       x = "Height (Inches)", y = "Count") +
  theme_minimal()
```




```
#mother's weight distribution
ggplot(babies_clean, aes(x = wt1)) +
  geom_histogram(bins = 30) +
  labs(title = "Mother's Weight Distribution",
       x = "Weight (Pounds)", y = "Count") +
  theme_minimal()
```

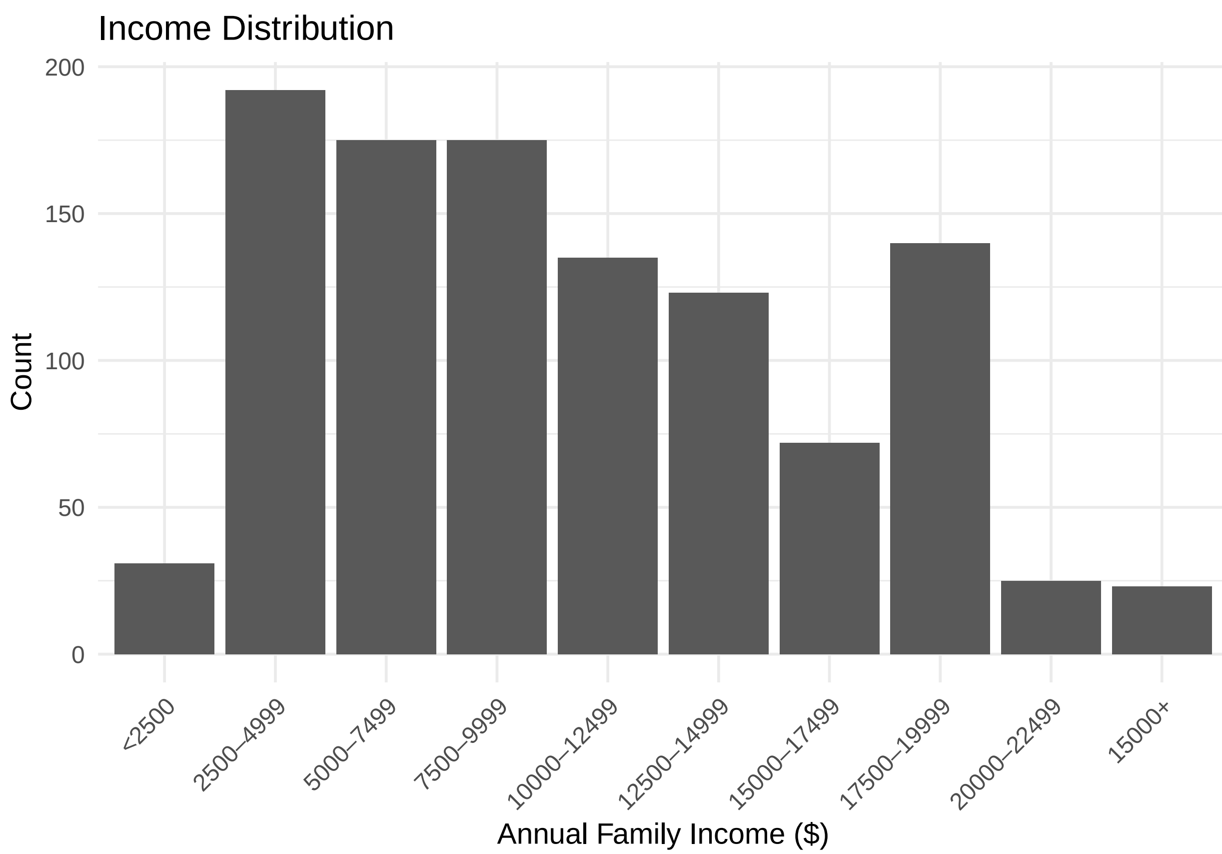



```r
#mother's race distribution
babies_clean$race_group <- with(babies_clean,
  ifelse(race %in% 0:5, "White",
  ifelse(race == 6, "Mexican",
  ifelse(race == 7, "Black",
  ifelse(race == 8, "Asian",
  ifelse(race == 9, "Mixed", NA))))))

babies_clean$race_group[babies_clean$race == 10] <- NA

ggplot(subset(babies_clean, !is.na(race_group)), aes(x = race_group)) +
  geom_bar() +
  labs(
    title = "Race Distribution",
    x = "Mother's Race",
    y = "Count"
  ) +
  theme_minimal()
```

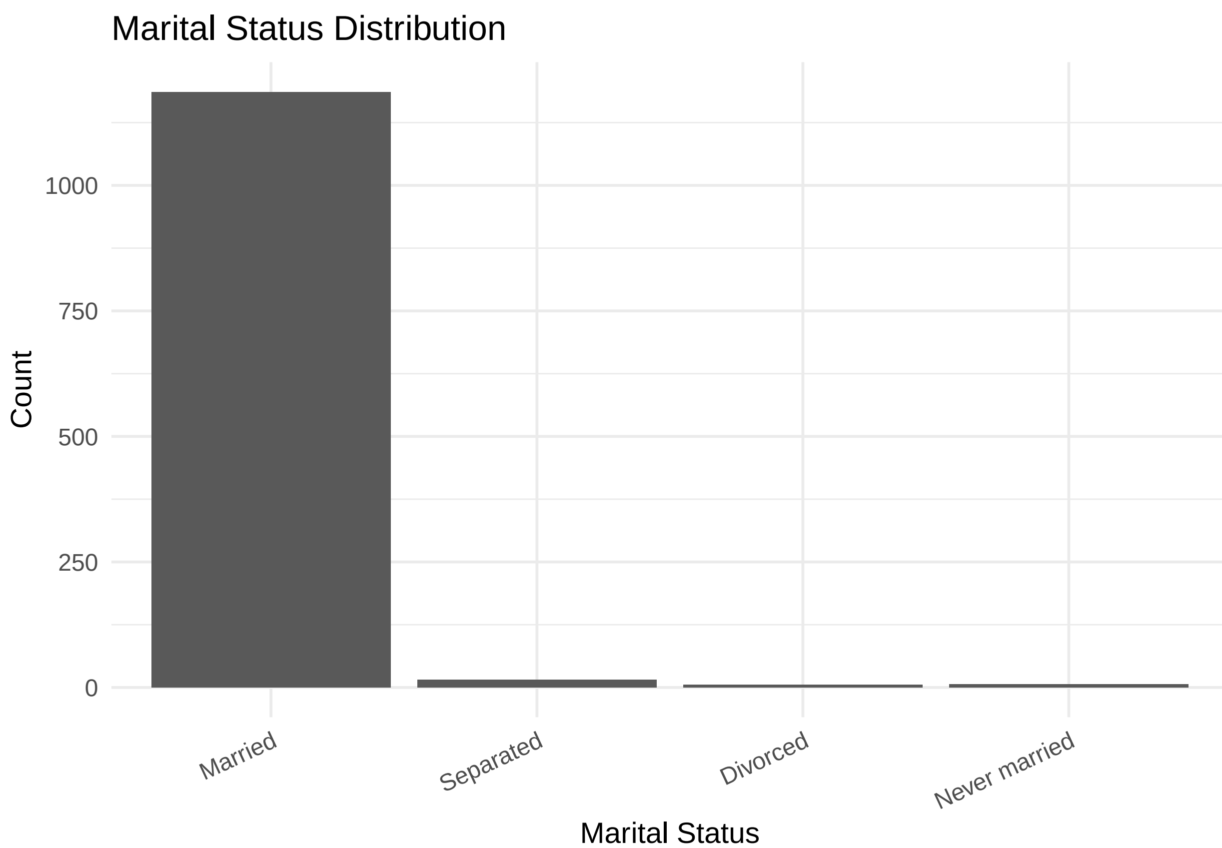



```
#annual income
babies_clean$inc_group <- factor(
  babies_clean$inc,
  levels = 0:9,
  labels = c(
    "<2500",
    "2500-4999",
    "5000-7499",
    "7500-9999",
    "10000-12499",
    "12500-14999",
    "15000-17499",
    "17500-19999",
    "20000-22499",
    "15000+"
  )
)

ggplot(subset(babies_clean, !is.na(inc_group)), aes(x = inc_group)) +
  geom_bar() +
  labs(
    title = "Income Distribution",
    x = "Annual Family Income ($)",
    y = "Count"
  ) +
  theme_minimal() +
```

```r
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

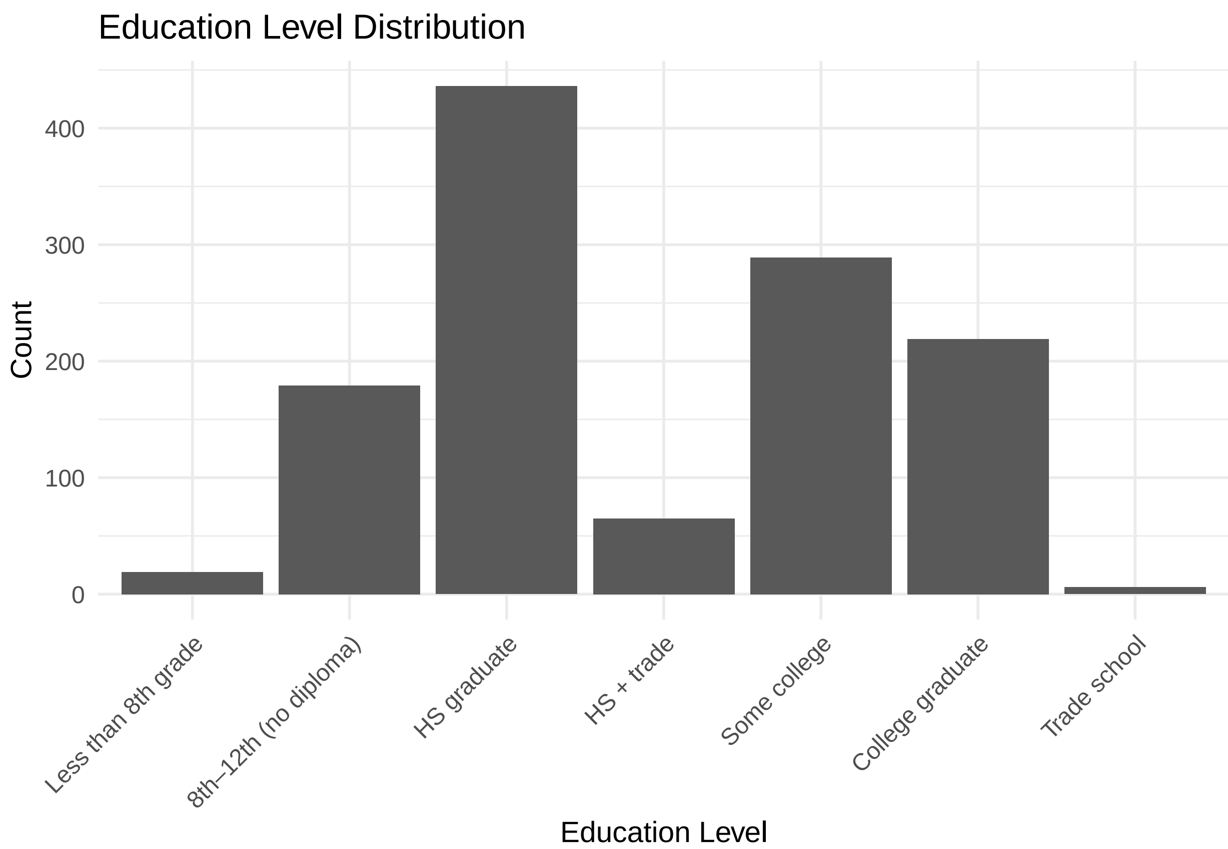



```r
#marital status distribution
babies_clean$marital_group <- factor(
  babies_clean$marital,
  levels = c(1,2,3,4,5),
  labels = c(
    "Married",
    "Separated",
    "Divorced",
    "Widowed",
    "Never married"
  )
)

ggplot(subset(babies_clean, !is.na(marital_group)), aes(x = marital_group)) +
  geom_bar() +
  labs(
    title = "Marital Status Distribution",
    x = "Marital Status",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
```




```
#mother's education distribution
babies_clean$edu_group <- with(babies_clean,
  ifelse(ed == 0, "Less than 8th grade",
  ifelse(ed == 1, "8th-12th (no diploma)",
  ifelse(ed == 2, "HS graduate",
  ifelse(ed == 3, "HS + trade",
  ifelse(ed == 4, "Some college",
  ifelse(ed == 5, "College graduate",
  ifelse(ed %in% c(6,7), "Trade school", NA))))))))

babies_clean$edu_group <- factor(
  babies_clean$edu_group,
  levels = c(
    "Less than 8th grade",
    "8th-12th (no diploma)",
    "HS graduate",
    "HS + trade",
    "Some college",
    "College graduate",
    "Trade school"
  )
)

ggplot(subset(babies_clean, !is.na(edu_group)), aes(x = edu_group)) +
  geom_bar() +
  labs(
```

```
    title = "Education Level Distribution",
    x = "Education Level",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



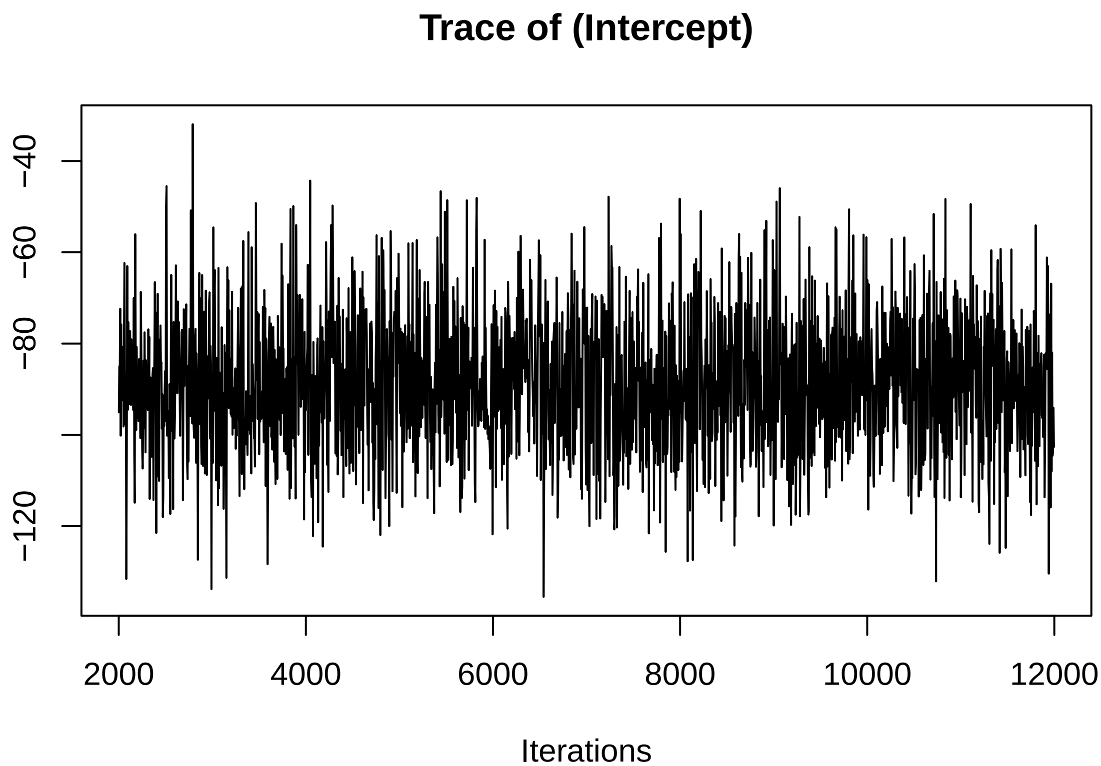
MCMC for Bayesian Linear Regression

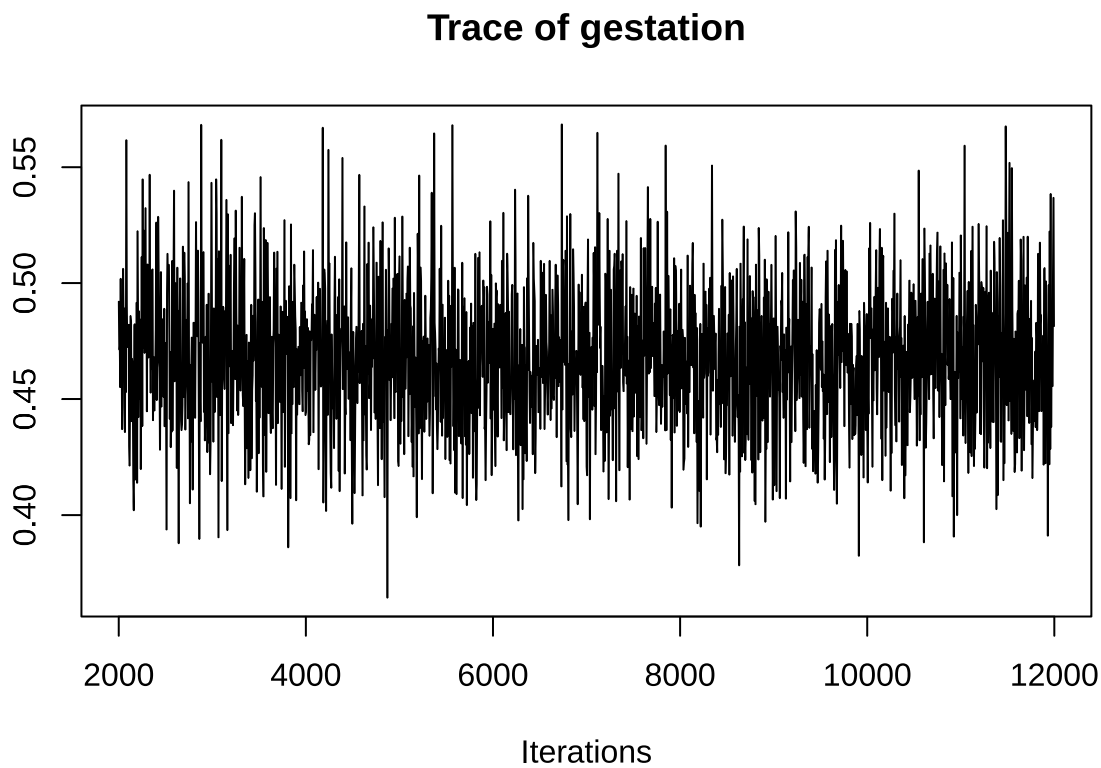
```
library(MCMCpack)

babies_clean$cigs_cat <- babies_clean$number

#remove unknowns
babies_clean$cigs_cat[babies_clean$cigs_cat %in% c(9, 98, 99)] <- NA

#convert to factor
babies_clean$cigs_cat <- factor(babies_clean$cigs_cat)

#linear regression
model <- MCMCregress(
  wt ~ gestation + ht + wt1 + number + parity + age,
  data = babies_clean,
```

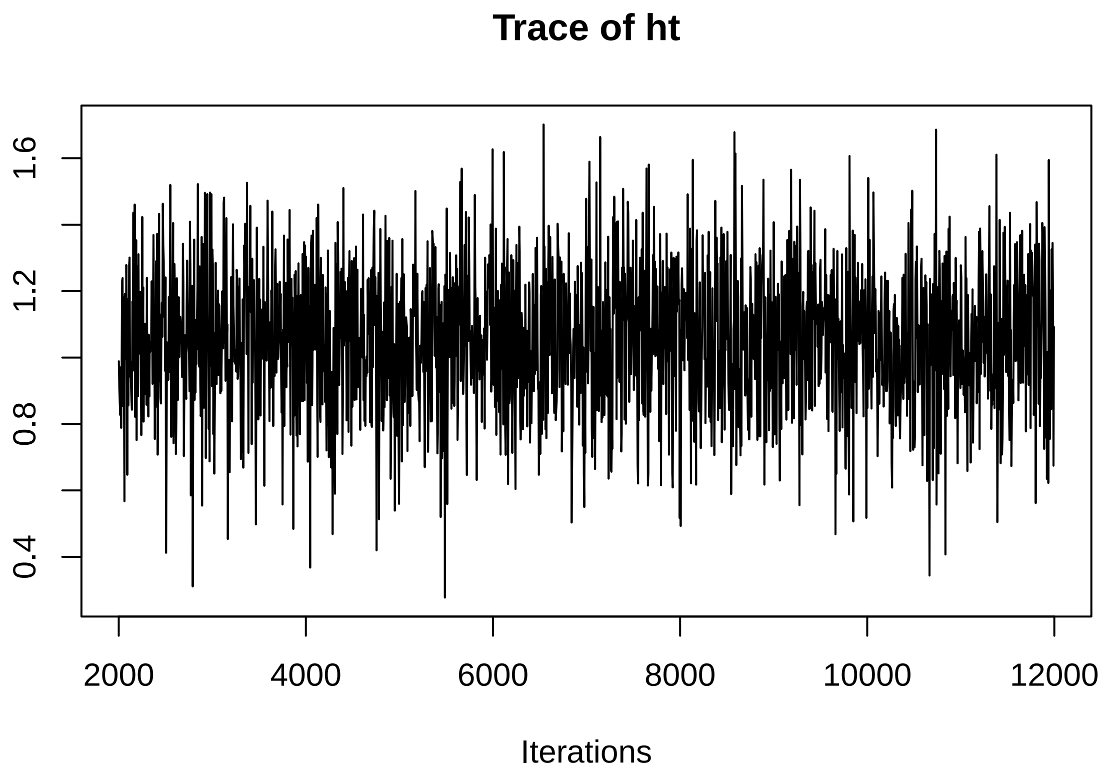
```r
  burnin = 2000,
  mcmc = 10000,
  thin = 5
)

summary(model)
```

```
##
## Iterations = 2001:11996
## Thinning interval = 5
## Number of chains = 1
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##                  Mean       SD  Naive SE Time-series SE
## (Intercept) -88.52626 14.97191 0.3347820      0.3026544
## gestation     0.46787  0.03056 0.0006832      0.0006832
## ht            1.05224  0.21078 0.0047132      0.0047132
## wt1           0.06318  0.02648 0.0005921      0.0005921
## number       -0.04559  0.05277 0.0011800      0.0011800
## parity        0.58461  0.30059 0.0067215      0.0067215
## age           0.03391  0.09632 0.0021537      0.0020974
## sigma2      268.56537 11.22891 0.2510861      0.2510861
##
## 2. Quantiles for each variable:
##
##                   2.5%       25%       50%       75%     97.5%
## (Intercept) -1.166e+02 -98.54726 -88.82171 -78.28579 -57.08025
## gestation    4.094e-01   0.44695   0.46757   0.48747   0.52823
## ht           6.344e-01   0.91043   1.05279   1.20036   1.45396
## wt1          1.303e-02   0.04504   0.06307   0.07997   0.11765
## number      -1.467e-01  -0.07993  -0.04537  -0.01025   0.06074
## parity       7.008e-03   0.38043   0.58144   0.78501   1.17836
## age         -1.544e-01  -0.03006   0.03153   0.10050   0.22076
## sigma2       2.481e+02 260.90331 268.29603 275.72618 291.76705
```

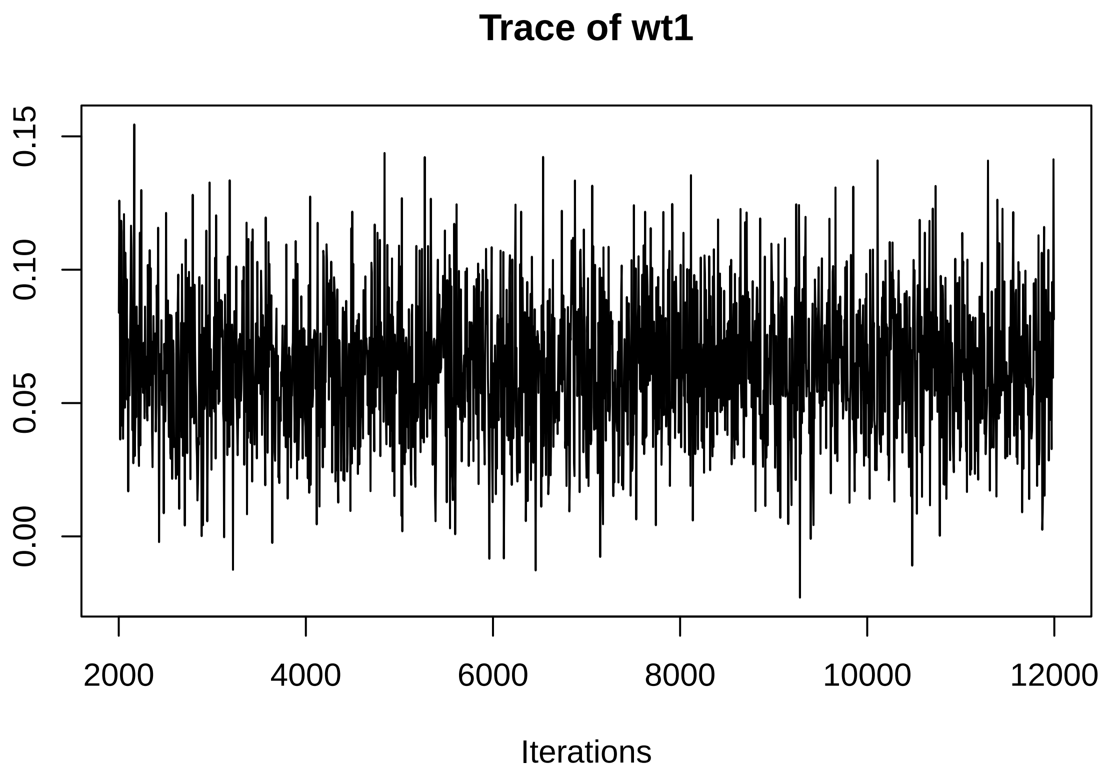
```r
#diagnostics
traceplot(model)
```

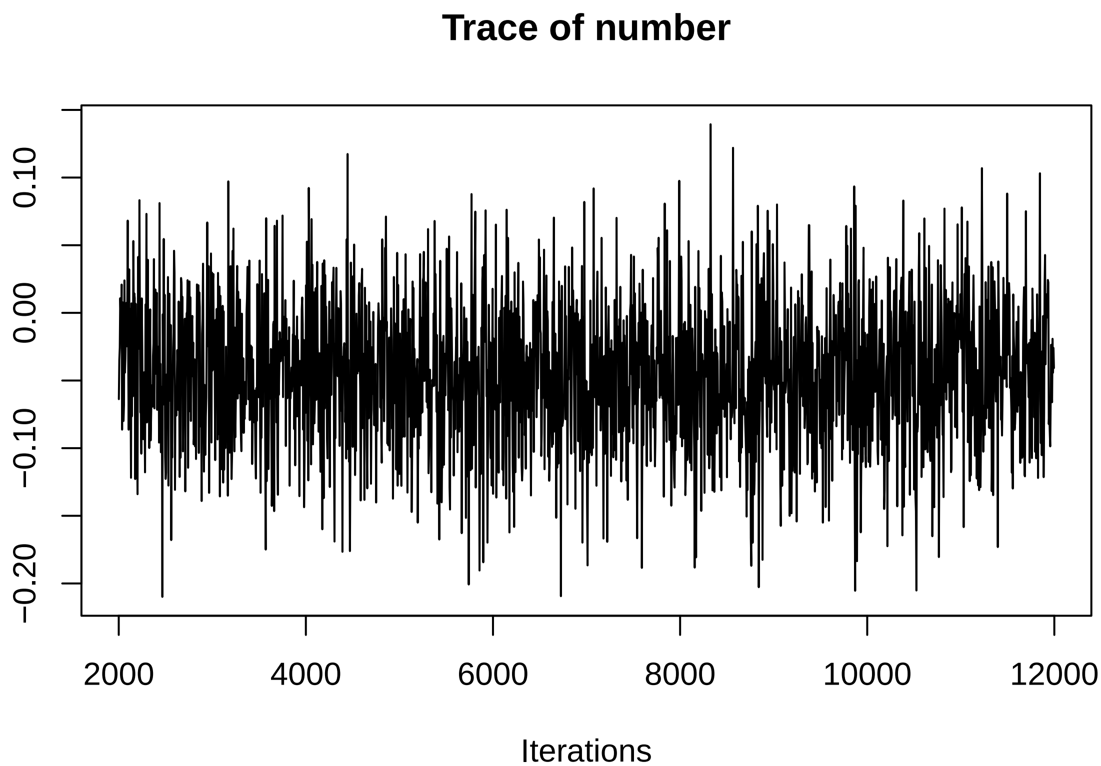

Trace of (Intercept)
−40
−60
−80
−120
2000
4000
6000
8000
10000
12000
Iterations

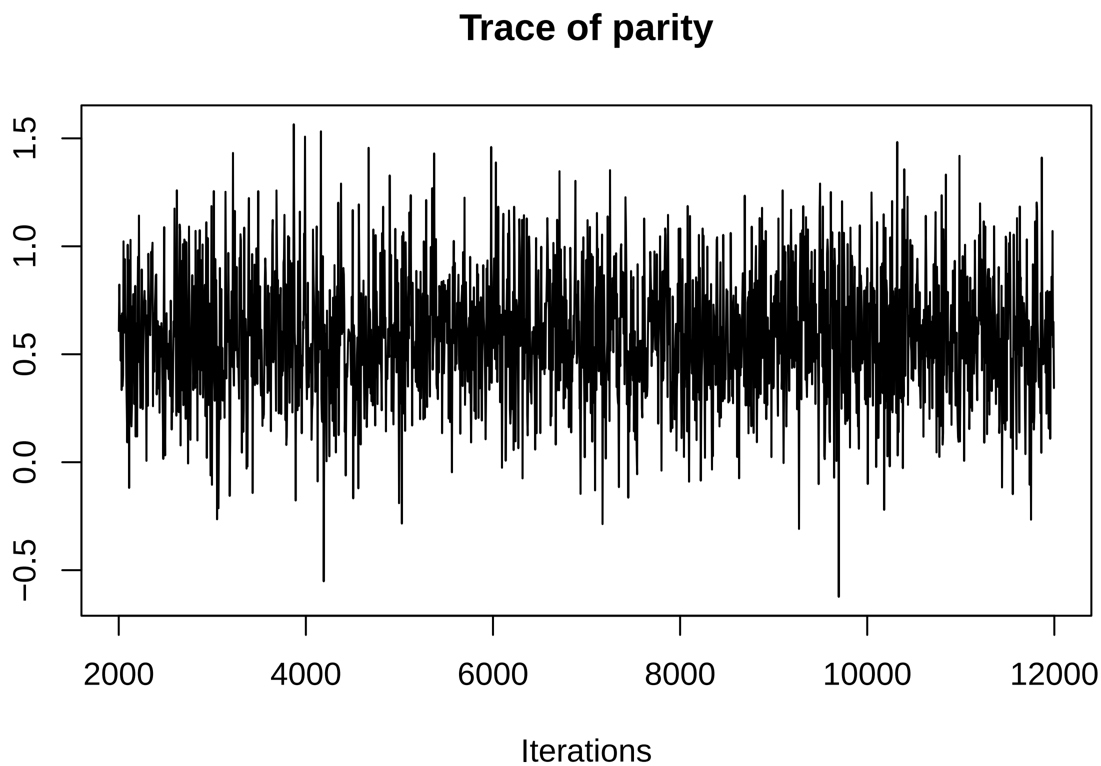

Trace of gestation
0.40
0.45
0.50
0.55
2000
4000
6000
8000
10000
12000
Iterations

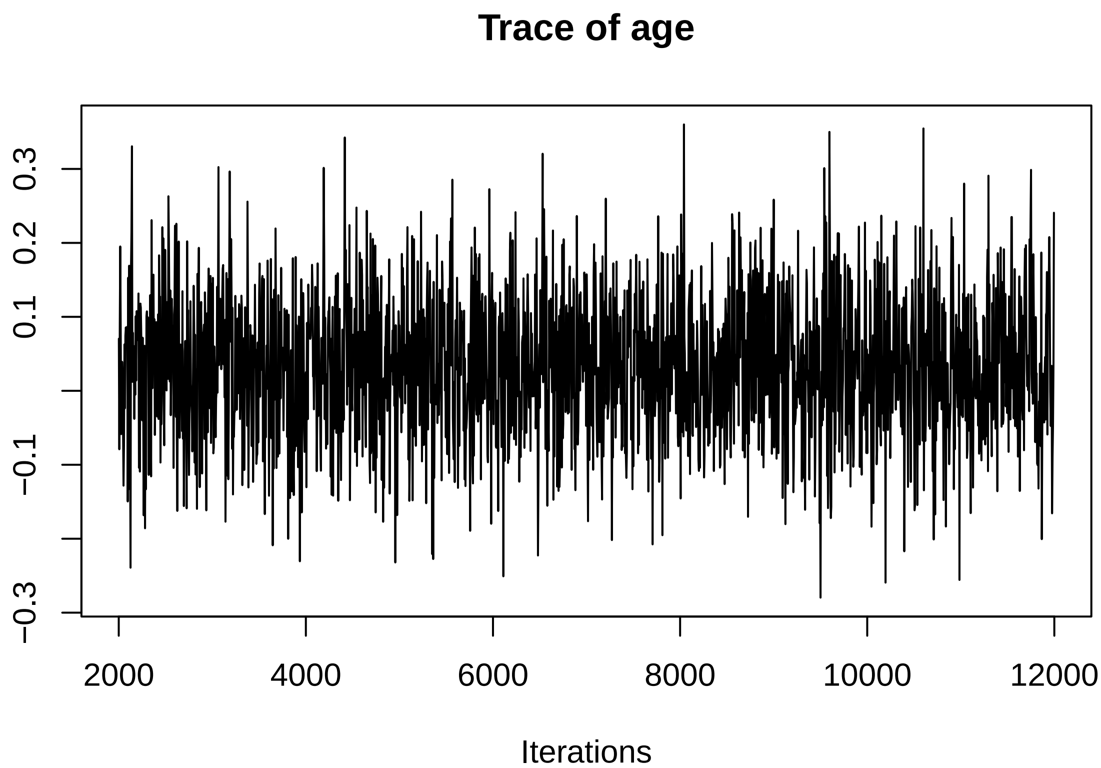


Trace of ht
1.6
1.2
0.8
0.4
2000
4000
6000
8000
10000
12000
Iterations

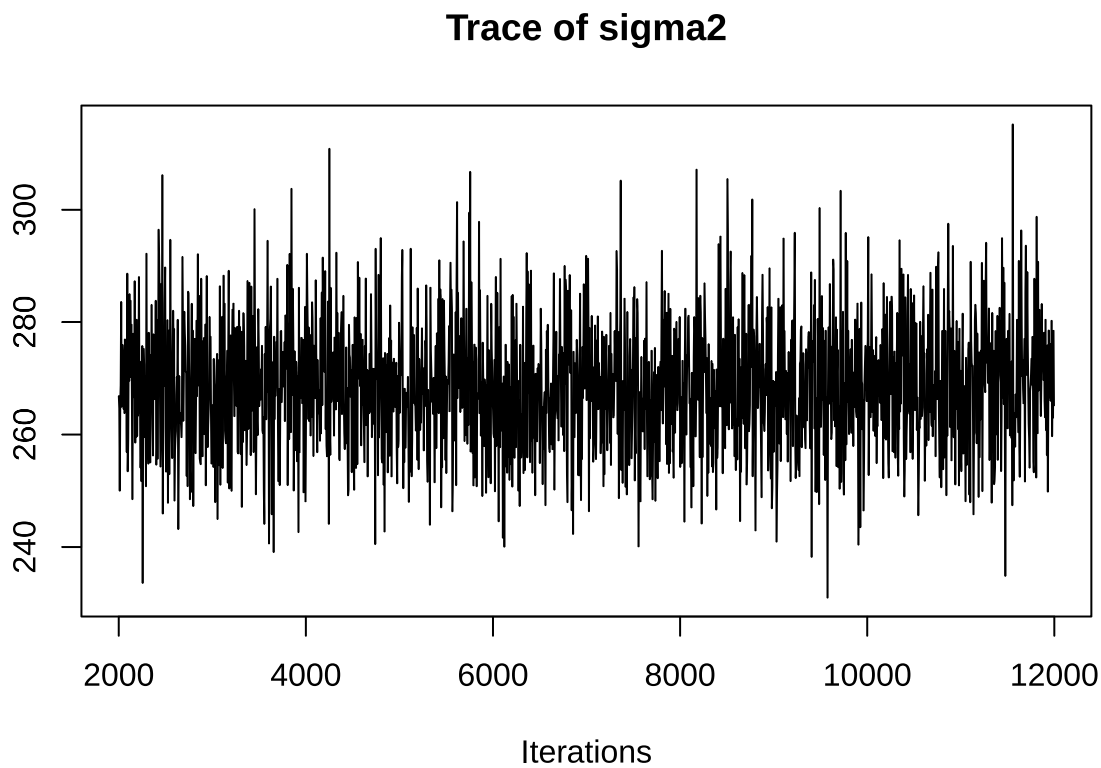

Trace of wt1
0.00
0.05
0.10
0.15
2000
4000
6000
8000
10000
12000
Iterations


Trace of number
0.10
0.00
−0.10
−0.20
2000
4000
6000
8000
10000
12000
Iterations

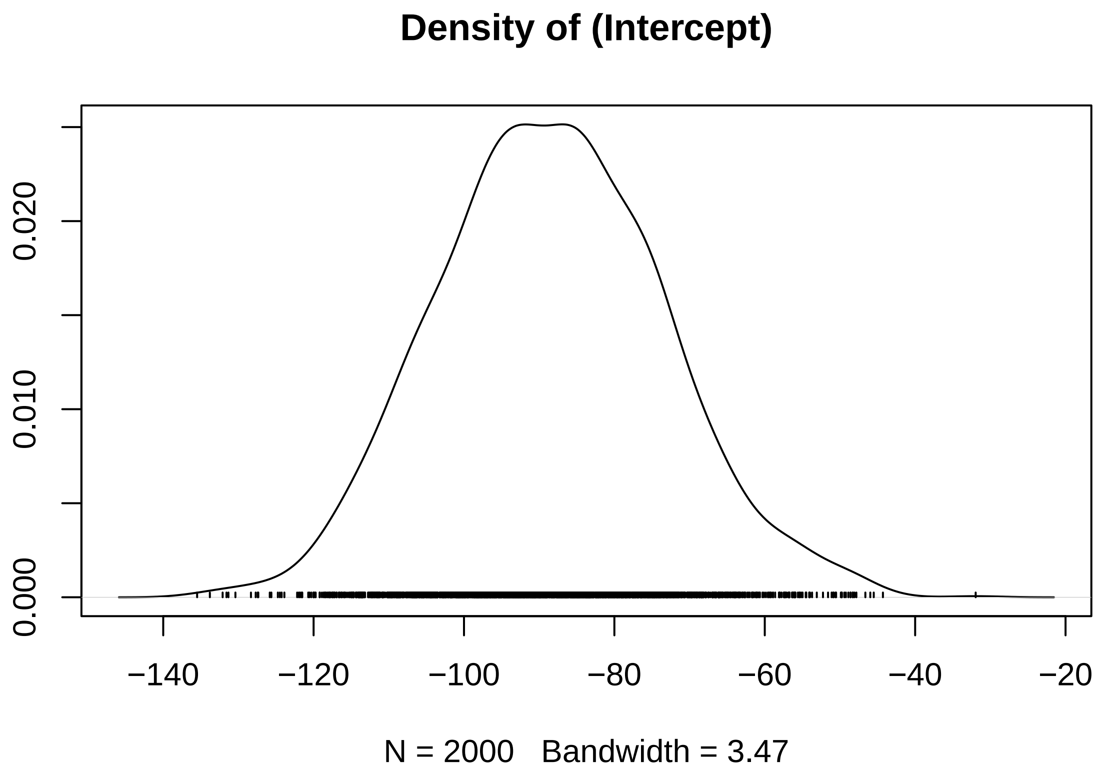

Trace of parity
1.5
1.0
0.5
0.0
−0.5
2000
4000
6000
8000
10000
12000
Iterations

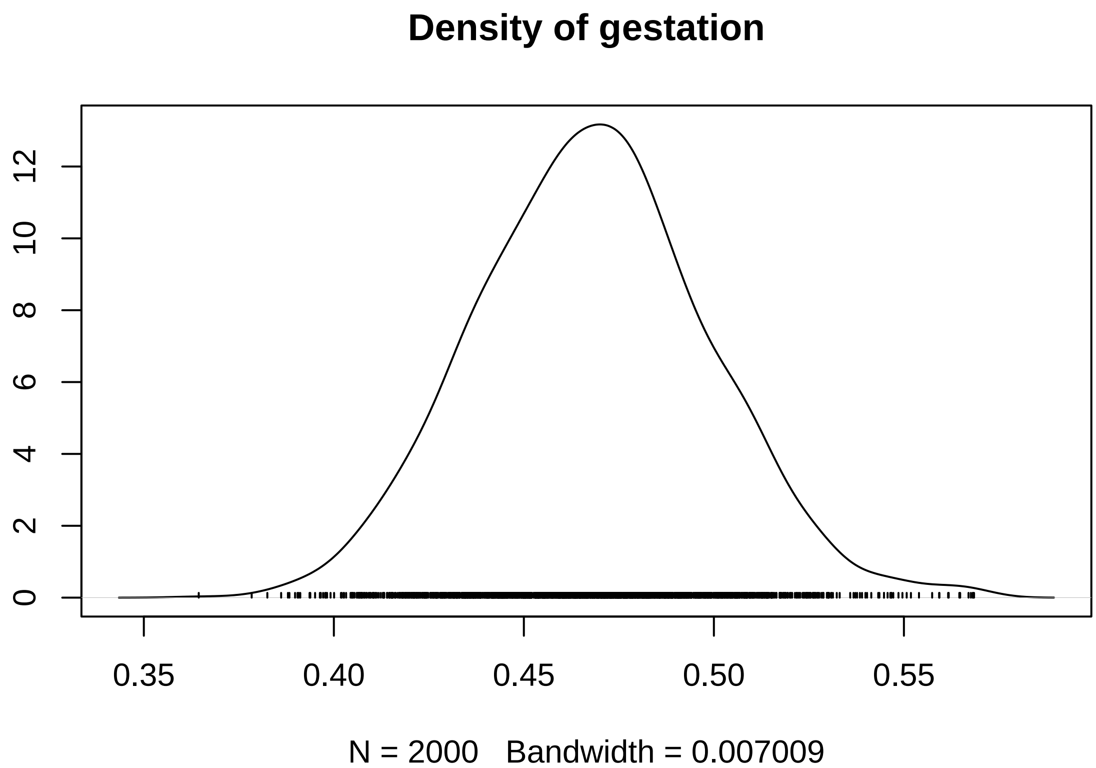


Trace of age
0.3
0.2
0.1
−0.1
−0.3
2000
4000
6000
8000
10000
12000
Iterations

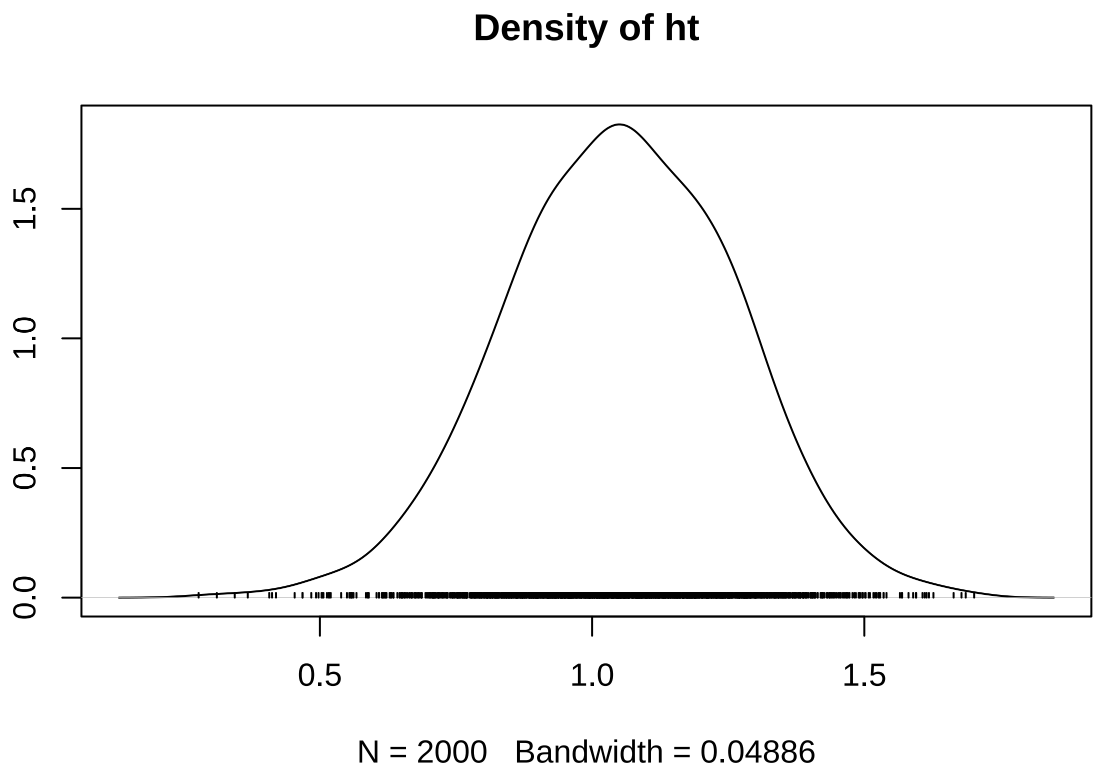



```r
#posterior distributions
densplot(model)
```

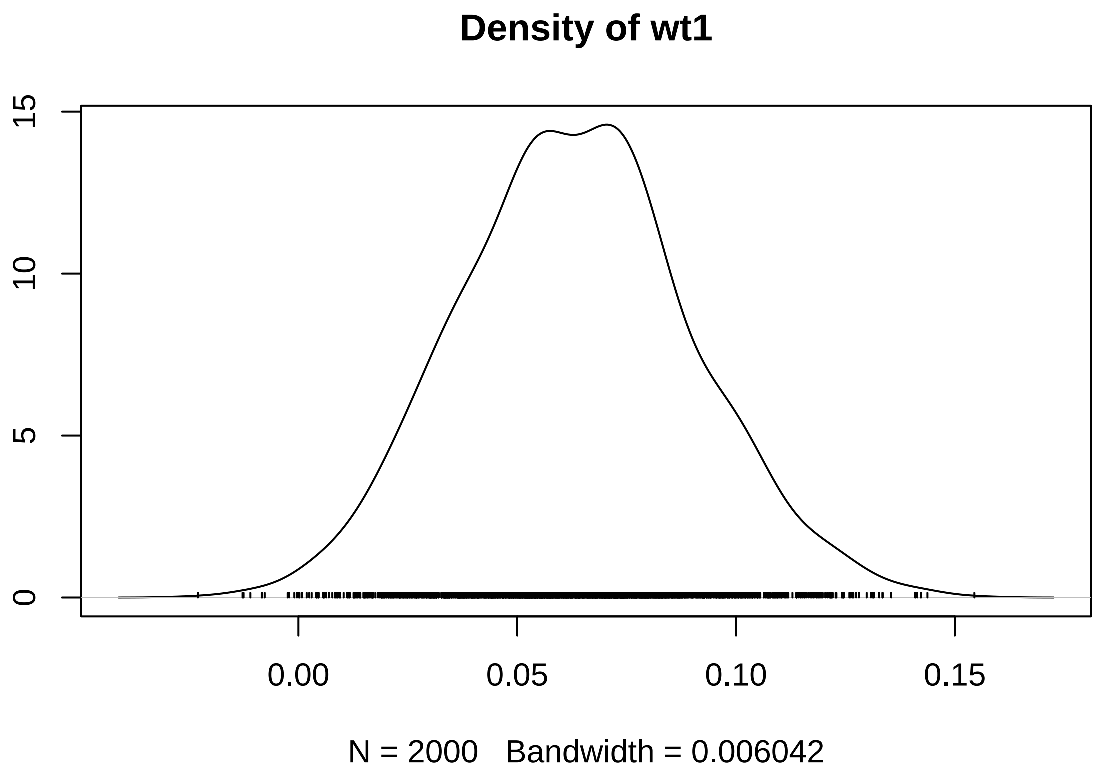

Density of (Intercept)
0.000
0.010
0.020
−140
−120
−100
−80
−60
−40
−20
N = 2000 Bandwidth = 3.47

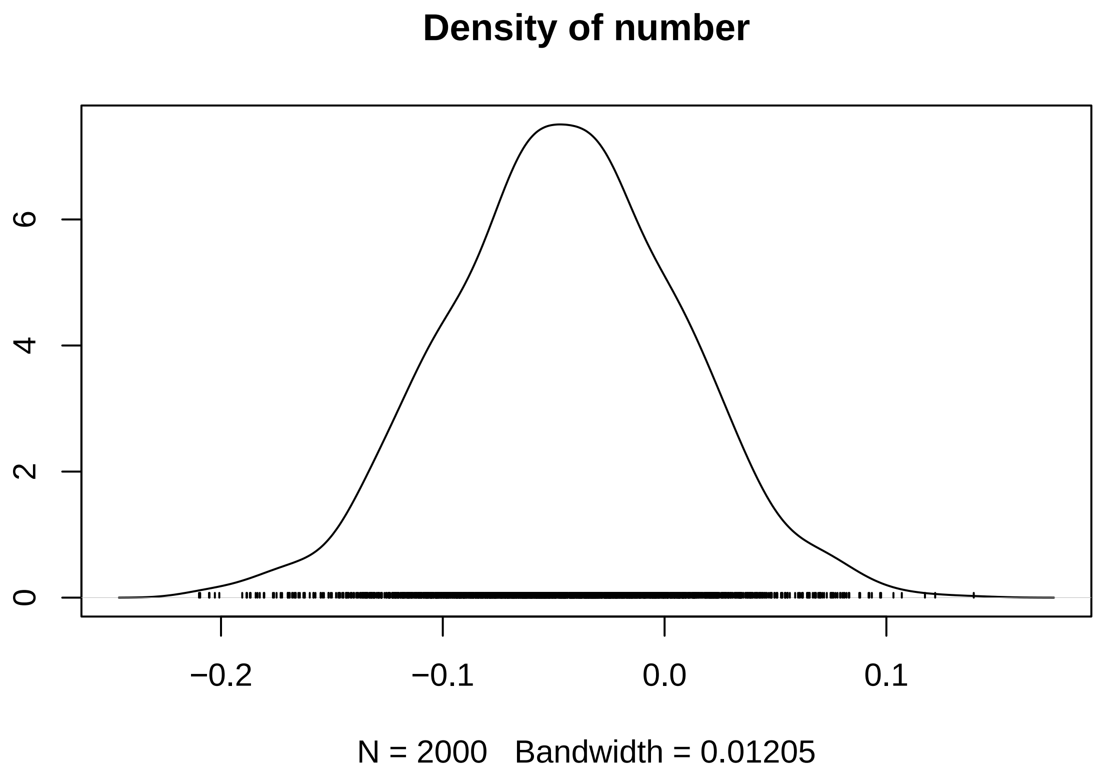


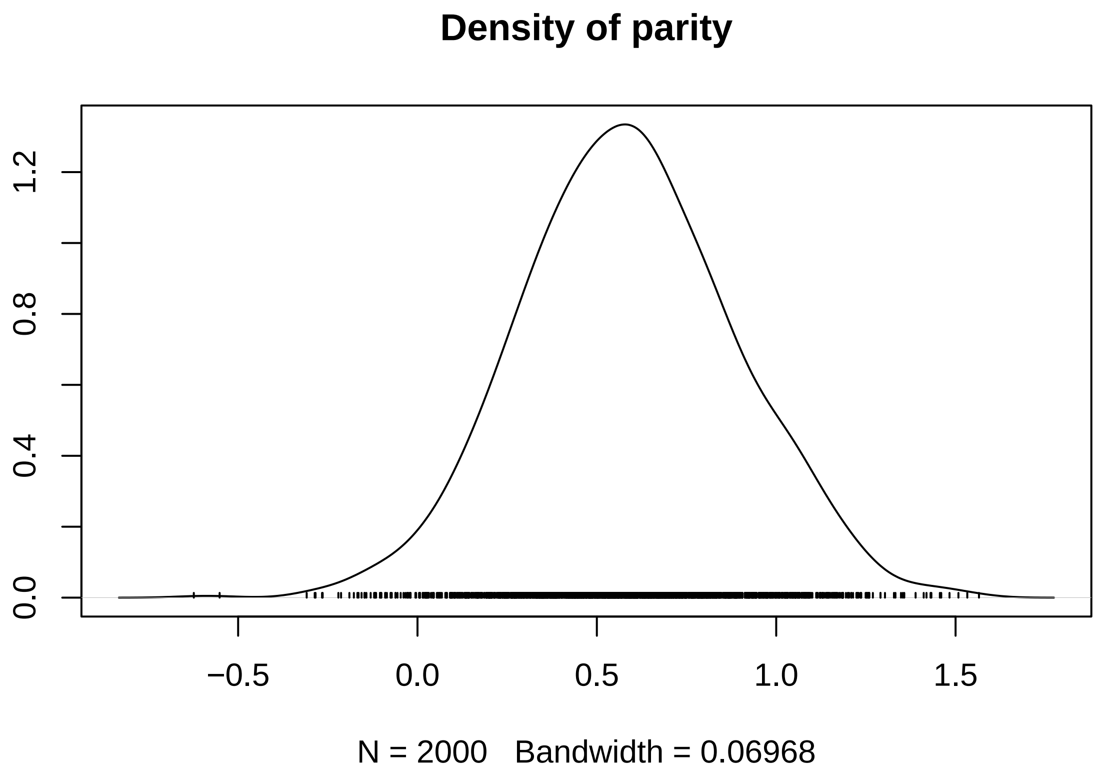
Density of gestation
0
2
4
6
8
10
12
0.35
0.40
0.45
0.50
0.55
N = 2000 Bandwidth = 0.007009



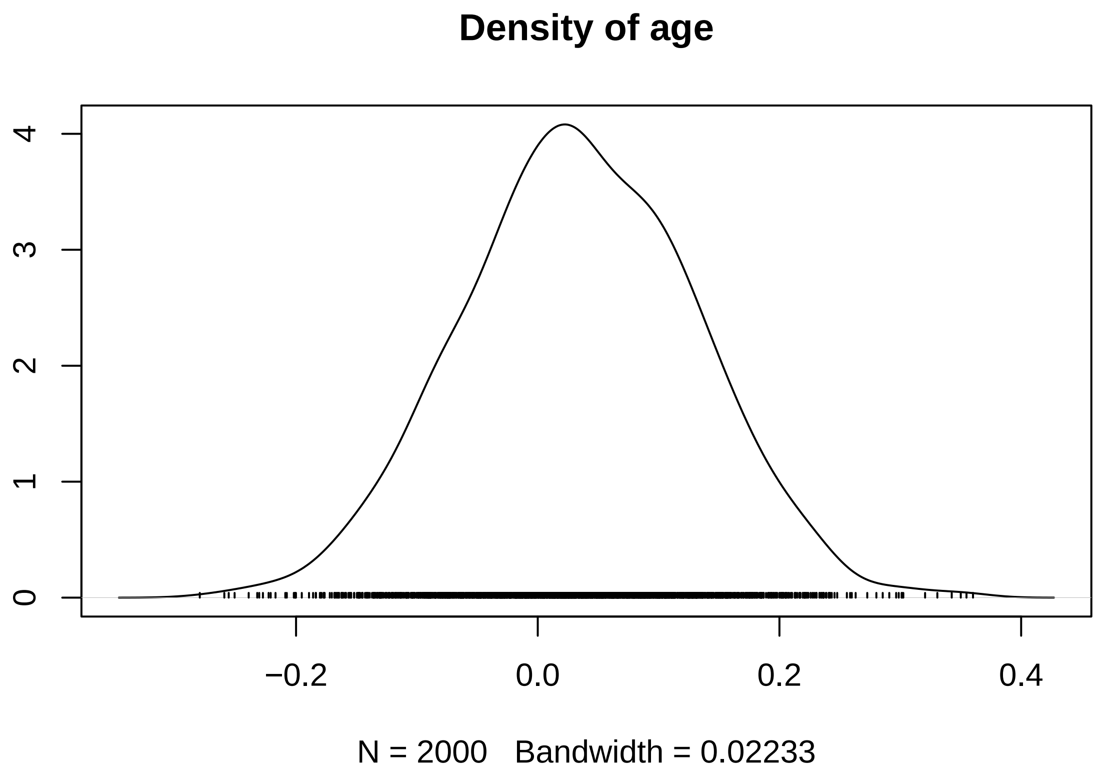
Density of ht
1.5
1.0
0.5
0.0
0.5
1.0
1.5
N = 2000 Bandwidth = 0.04886

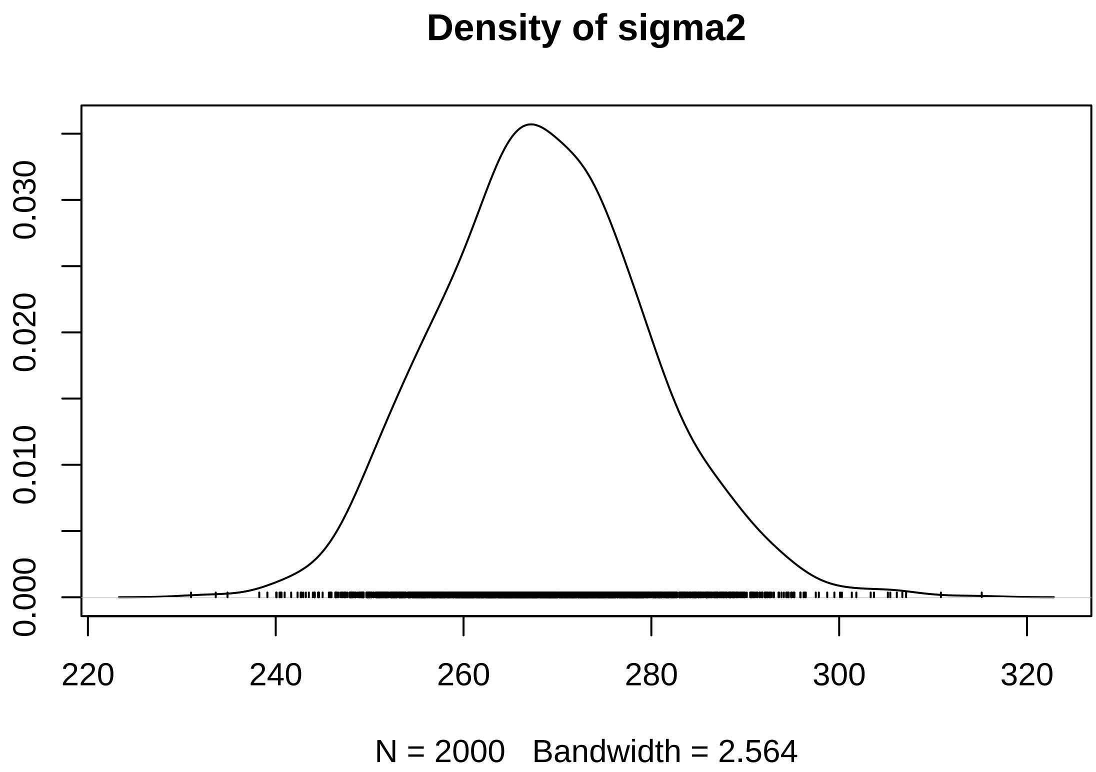

Density of wt1
0
5
10
15
0.00
0.05
0.10
0.15
N = 2000 Bandwidth = 0.006042


Density of number
0
2
4
6
−0.2
−0.1
0.0
0.1
N = 2000   Bandwidth = 0.01205

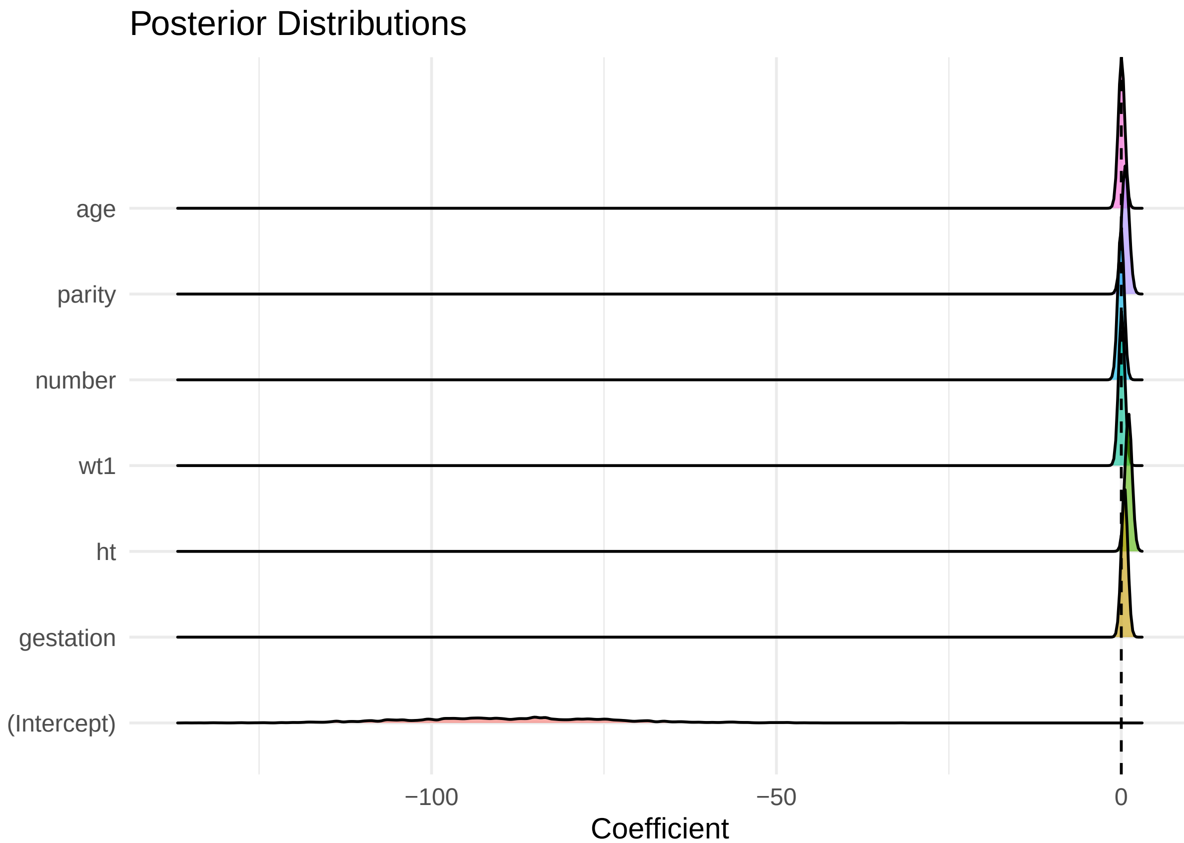


Density of parity
1.2
0.8
0.4
0.0
−0.5
0.0
0.5
1.0
1.5
N = 2000   Bandwidth = 0.06968


Density of age
4
3
2
1
0
−0.2
0.0
0.2
0.4
N = 2000   Bandwidth = 0.02233




```r
#ridgeline plot
m <- as.matrix(model)

#convert all of coefficients to dataframe
df <- as.data.frame(m)
df <- df[, !colnames(df) %in% "sigma2"]
df_long <- melt(df)

#plot
ggplot(df_long, aes(x = value, y = variable, fill = variable)) +
  geom_density_ridges(alpha = 0.6) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(title = "Posterior Distributions",
       x = "Coefficient", y = "") +
  theme_minimal() +
  theme(legend.position = "none")
```


Posterior Distributions
age
parity
number
wt1
ht
gestation
(Intercept)
−100
−50
0
Coefficient